RIDWAN AHMED

Paris, France | +33 7 53 07 77 02 | ridwan.ahmed.auditeur@lecnam.net

rid1-portfolio.pages.dev | linkedin.com/in/rid1ahmed

PROFESSIONAL PROFILE

Mechanical engineer pursuing the International Master in Structural Mechanics and Coupled Systems (M2) at Cnam Paris. Published author with experimental aluminium-matrix composite research and practical experience in numerical modelling, finite-element analysis, CAD and scientific programming. Seeking a research-oriented internship to address real-world engineering challenges across related disciplines.

EDUCATION

International Master in Structural Mechanics and Coupled Systems (M2) | 2025–2027 (expected)
Conservatoire national des arts et métiers (Cnam), Paris, France

B.Sc. Engg. in Mechanical Engineering | 2023
Bangladesh Army University of Science and Technology (BAUST), Bangladesh

RESEARCH AND PUBLICATION

Published author. Ahmed, R. et al. “Effect of jute stick charcoal powder on aluminum metal matrix composite characterization.” *Results in Engineering* (Elsevier), 2024. doi:10.1016/j.rineng.2024.101990.

Conducted composite fabrication and experimental characterization involving stir casting, mechanical testing, scanning electron microscopy (SEM) and X-ray diffraction (XRD).

PROFESSIONAL EXPERIENCE

Data Development Lead | 2025
Takeoff Moving Company, Canada (remote)

- Developed operational data workflows and supported website/data processes.

Executive, Operations & Supply Chain | 2024
Esquire Group Limited, Bangladesh

- Supported manufacturing operations, maintenance coordination and process improvement.

Assistant Engineer | 2023–2024
PEAL Engineering & Construction Ltd., Bangladesh

- Assisted HDD project execution, procurement, tendering and technical documentation.

Industrial Attaché | 2022
Central Locomotive Workshop, Bangladesh Railway

- Observed locomotive maintenance and repair operations.

Engineering Intern | 2021
Battery Low Interactive Ltd., Bangladesh

- Contributed to quality assurance and testing of web and VR applications.

TECHNICAL SKILLS

Numerical simulation: ANSYS, MSC Patran/Nastran, COMSOL, FEA, structural analysis

Design and CAD: SolidWorks, AutoCAD, mechanical design and wind-tunnel CAD

Programming and analysis: Python, MATLAB, numerical methods, scientific programming, Excel

Experimental and embedded: Composite fabrication, SEM, mechanical testing, microcontrollers and sensors

SELECTED PROJECTS

- Wind-turbine blade notch effects — finite-element investigation of stress concentrations and structural behaviour.
- Jute-stick-charcoal aluminium-matrix composites — fabrication and mechanical characterization for peer-reviewed research.
- Open-channel desktop wind tunnel — CAD design and practical aerodynamic testing setup.

- Numerical methods and modelling — finite-element implementation and optimisation exercises using MATLAB and Python.

LEADERSHIP, AWARDS AND EXTRACURRICULAR ACTIVITIES

- President, BAUST Cultural Society (5th Committee); Prefect, BNMPC.
- NASA Space Apps Collective (2023) — selected among 30 global Space Leaders.
- Winner, GPH Ispat “Esho Robot Banai” Idea Contest; participant in 72-hour hackathon.
- Runner-up, BAUST intra-university instrumental music competition.
- Delegate (Iran), Economic and Financial Committee, Bangladesh Model United Nations (2016).
- Participant, SustainX Global Case Study Competition (2021); Microsoft Engineering Virtual Experience Program.
- Member, International Association of Engineers (IAENG); participant, Unity VR game development workshop.
- Peer reviewer, Elsevier journal Results in Engineering.

LANGUAGES

Bangla (native) | English (IELTS Academic 8.0) | French (A2)

ACADEMIC REFERENCES

Lucie Rouleau — Maître de conférences–HDR, LMSSC, Cnam Paris | lucie.rouleau@lecnam.net

Jean-François Deü — Professor, LMSSC, Cnam Paris | jean-francois.deu@cnam.fr